1. Item 6

> Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

This problem is set in a real-world context (making salad dressing). Students need to identify the mathematical elements (proportional relationships) and key variables (the volumes of salad oil, vinegar, soy sauce, and their ratios).

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2. Item 4

> Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

Students must read and understand the data in the recipe table, clarify the proportional relationships between the ingredients, and build a proportional model for a new total volume (150 mL).

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3. Item 7

> Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

The problem simplifies the types and proportions of ingredients. Students can decompose it into a proportional calculation problem, ignoring other real-world factors (like mixing, dissolving, etc.).

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4. Item 9

> Translate a problem into a standard mathematical representation or algorithm

Students need to translate "making 150 mL of dressing" into a proportional calculation problem, using an algorithm such as:

$$\text{Salad oil needed} = \frac{60}{100} \times 150$$

5. Item 5

> Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

This problem contains the underlying pattern of "fixed ingredient proportions." Students must recognize this structure and apply it to solve the problem.

6. Item 10

> Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

Although not explicitly required, students can use proportions, fractions, or multiplication to represent the relationships, for example:

$$60 : 30 : 10 = 6 : 3 : 1$$

and then calculate the amounts for the new total volume.

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7. Item 1

> Select a mathematical description or a representation that describes a problem

Students need to choose "proportion" or "fraction" as the primary mathematical tool to describe the problem.

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8. Item 8

> Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

At its core, this is a **proportional scaling problem**. Students should connect it to previously learned concepts such as "ratio allocation" or "unit rate" calculation methods.

1. 第 6 条

> Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

本题是一道实际情境问题（制作沙拉酱），学生需要从中识别出数学要素（比例关系）和关键变量（沙拉油、醋、酱油的体积及其比例）。

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2. 第 4 条

> Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

学生需要阅读并理解食谱表格中的数据，理清各成分的比例关系，并在新的总量（150 mL）下建立比例模型。

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3. 第 7 条

> Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

题目已经简化了调味品的种类和比例，学生可以将其分解为比例计算问题，忽略实际制作中的其他因素（如搅拌、溶解等）。

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4. 第 9 条

> Translate a problem into a standard mathematical representation or algorithm

学生需要将“制作150 mL酱料”转化为比例计算问题，如使用以下算法：

$$\text{沙拉油用量} = \frac{60}{100} \times 150$$

5. 第 5 条

> Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

本题中隐藏着“各成分比例不变”的规律，学生需要识别出这一结构，并用于解题。

6. 第 10 条

> Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

虽然没有明确要求画图或列方程，但学生可以用比例式、分数或乘法来表示关系，例如：

$$60 : 30 : 10 = 6 : 3 : 1$$

并推算新总量下各成分的量。

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7. 第 1 条

> Select a mathematical description or a representation that describes a problem

学生需要选择“比例”或“分数”作为描述问题的主要数学工具。

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8. 第 8 条

> Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

这题本质上是一个**比例缩放问题**，学生需要联想到已学过的“按比例分配”或“单位比例”计算方法。